PLSC 30600

Week 6: Design-based inference

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Loading packages for this class

```
> # install.packages('../assets/ri_0.9.tar.gz',
> #                  repos = NULL, type = 'source')
> library(ri)
> library(ggplot2)
> set.seed(60637)
```

Example adapted from:

Chattopadhyay, Raghabendra & Duflo, Esther (2004). Women as policy makers: Evidence from a randomized policy experiment in India.

as discussed in

Gerber, Alan S. & Green, Donald P. (2012). Field experiments: Design, analysis, and interpretation.

Chattopadhyay & Duflo example

- Population: 7 villages

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	$Y_i(0)$	$Y_i(1)$	D_i	Y_i	τ_i
Village 1	10	15	1	15	5
Village 2	15	15	0	15	0
Village 3	20	30	0	20	10
Village 4	20	15	0	20	-5
Village 5	10	20	0	10	10
Village 6	15	15	0	15	0
Village 7	15	30	1	30	15
Average	**15**	**20**			**5**

$$\text{ATE} = \mathrm{E}\left[\tau_i\right] = 5$$

Chattopadhyay & Duflo example

What we actually see:

	$Y_i(0)$	$Y_i(1)$	D_i	Y_i	τ_i
Village 1	?	15	1	15	?
Village 2	15	?	0	15	?
Village 3	20	?	0	20	?
Village 4	20	?	0	20	?
Village 5	10	?	0	10	?
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Village 5	10	?	0	10	?
Village 6	15	?	0	15	?
Village 7	?	30	1	30	?

To produce an *estimate* of the ATE, we can compare people who received treatment 1 to people who received treatment 0.

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Village 6	15	?	0	15	?
Village 7	?	30	1	30	?

To produce an *estimate* of the ATE, we can compare people who received treatment 1 to people who received treatment 0.

$$\bar{Y}(1) = 22.5 \qquad \bar{Y}(0) = 16$$

Chattopadhyay & Duflo example

What we actually see:

	$Y_i(0)$	$Y_i(1)$	D_i	Y_i	τ_i
Village 1	?	15	1	15	?
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Village 3	20	?	0	20	?
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Village 5	10	?	0	10	?
Village 6	15	?	0	15	?
Village 7	?	30	1	30	?

To produce an *estimate* of the ATE, we can compare people who received treatment 1 to people who received treatment 0.

$$\bar{Y}(1) = 22.5 \qquad \bar{Y}(0) = 16$$

$$\hat{\tau} = 22.5 - 16 = 6.5$$

estimand

estimate

Ingredients

- 150g unsalted butter, plus extra for greasing
- 150g plain chocolate, broken into pieces
- 150g plain flour
- ½ tsp baking powder
- ½ tsp bicarbonate of soda
- 200g light muscovado sugar
- 2 large eggs

Method

1. Heat the oven to 160C/140C fan/gas 3. Grease and base line a 1 litre heatproof glass pudding basin and a 450g loaf tin with baking parchment.

2. Put the butter and chocolate into a saucepan and melt over a low heat, stirring. When the chocolate has all melted remove from the heat.

estimator

(image cred: simongrund89)

Difference-in-means

Proposed estimator for $\mathrm{E}[\tau_i]$: compare people who received treatment 1 to people who received treatment 0.

$$\hat{\tau}_{DM} = \frac{\sum_i^n Y_i D_i}{\sum_i^n D_i} - \frac{\sum_i^n Y_i (1 - D_i)}{\sum_i^n (1 - D_i)}$$

Chattopadhyay & Duflo example

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Chattopadhyay & Duflo example

- We have produced an estimate here, by taking the difference in means.
- But what can we say about our uncertainty about this estimate? The number of observations in each group is pretty small.
- Is the estimate meaningfully different from zero?
- How likely would we be to see an effect this size just by chance?

Chattopadhyay & Duflo example

- One way to think about this is to assume the individual treatment effect for all individuals is exactly zero. Then, no matter how we randomized treatment assignment, we would see the same Y_is.

Chattopadhyay & Duflo example

- One way to think about this is to assume the individual treatment effect for all individuals is exactly zero. Then, no matter how we randomized treatment assignment, we would see the same Y_is.
- We can then say how often we would see a treatment effect estimate of this size just by chance, under the assumption that individual treatment effects were actually zero.

Sharp null hypothesis of no effect

Assuming all treatment effects are exactly zero is called the **sharp null hypothesis of no effect.**

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Assuming all treatment effects are exactly zero is called the **sharp null hypothesis of no effect.** Also referred to as "Fisher's null" after Sir Ronald Aylmer Fisher (1890-1962).

Sharp null hypothesis of no effect

We might write the sharp null hypothesis this way:

$$\mathrm{H}_0 : \tau_i = 0, \text{ for all } i \text{ in our pop.}$$

This implies that potential outcomes are identical under treatment and control, for all individuals.

$$Y_i(0) = Y_i(1)$$

Sharp null hypothesis of no effect

- To test this hypothesis, we also need to know (or assume we know) the randomization procedure

Sharp null hypothesis of no effect

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- Here, we'll assume exactly two villages are assigned treatment.

Chattopadhyay & Duflo example

Then we can fill in this table…

	$Y_i(0)$	$Y_i(1)$	D_i	Y_i	τ_i
Village 1	?	15	1	15	?
Village 2	15	?	0	15	?
Village 3	20	?	0	20	?
Village 4	20	?	0	20	?
Village 5	10	?	0	10	?
Village 6	15	?	0	15	?
Village 7	?	30	1	30	?

Chattopadhyay & Duflo example

…as this:

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Village 6	15	15	0	15	0
Village 7	30	30	1	30	0

Gerber Green example

We can re-run the randomization, and the potential outcomes and observed Y_i will not change, but the treatment effect estimate will.

	$Y_i(0)$	$Y_i(1)$	D_i	Y_i	τ_i
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Village 5	10	10	0	10	0
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Village 7	30	30	0	30	0

$$\bar{Y}(1) = 15 \qquad \bar{Y}(0) = 19$$

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Village 5	10	10	0	10	0
Village 6	15	15	1	15	0
Village 7	30	30	0	30	0

$$\bar{Y}(1) = 15 \qquad \bar{Y}(0) = 19$$

$$\hat{\tau} = 15 - 19 = -4$$

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Village 6	15	15	0	15	0
Village 7	30	30	0	30	0

$$\bar{Y}(1) = 20 \qquad \bar{Y}(0) = 17$$

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Village 6	15	15	0	15	0
Village 7	30	30	0	30	0

$$\bar{Y}(1) = 20 \qquad \bar{Y}(0) = 17$$

$$\hat{\tau} = 20 - 17 = 3$$

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Village 3	20	20	1	20	0
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Village 6	15	15	0	15	0
Village 7	30	30	1	30	0

$$\bar{Y}(1) = 25 \qquad \bar{Y}(0) = 15$$

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	$Y_i(0)$	$Y_i(1)$	D_i	Y_i	τ_i
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Village 4	20	20	0	20	0
Village 5	10	10	0	10	0
Village 6	15	15	0	15	0
Village 7	30	30	1	30	0

$$\bar{Y}(1) = 25 \qquad \bar{Y}(0) = 15$$

$$\hat{\tau} = 25 - 15 = 10$$

Randomization procedure

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$$\Pr(\boldsymbol{D} = \boldsymbol{d}) = \frac{1}{\binom{7}{2}} = \frac{1}{21}.$$

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- Each assignment vector $\boldsymbol{D}$ with $\sum_i D_i = 2$ is equally likely:

$$\Pr(\boldsymbol{D} = \boldsymbol{d}) = \frac{1}{\binom{7}{2}} = \frac{1}{21}.$$

- The marginal assignment probability is the same for each village:

$$\pi_i \equiv \Pr(D_i = 1) = \frac{2}{7}.$$

We can use the package `ri` to find the *exact* distribution of $\hat{\tau}_{DM}$ under the sharp null.

Our real data:

```
> df <- data.frame(
+         # our initial treatment vector
+         D = c(1, 0, 0, 0, 0, 0, 1),
+         # our initial response vector
+         Y = c(15, 15, 20, 20, 10, 15, 30),
+         # marginal assignment probability: pi_i = Pr(D_i = 1)
+         pi = rep(2/7, 7)
+ )
> df
```

```
  D  Y        pi
1 1 15 0.2857143
2 0 15 0.2857143
3 0 20 0.2857143
4 0 20 0.2857143
5 0 10 0.2857143
6 0 15 0.2857143
7 1 30 0.2857143
```

And our difference in means estimate of the average treatment effect under the real data:

```
> Y1 <- df$Y[which(df$D == 1)]
> Y0 <- df$Y[which(df$D == 0)]
> (dm_hat <- mean(Y1) - mean(Y0))
[1] 6.5
```

Adding in the hypothetical data.

```
> df <-  cbind( # binds the columns together
+          df,
+          # Y(0) under the sharp null of no effect
+          Y0 = df$Y,
+          # Y(1) under the sharp null of no effect
+          Y1 = df$Y)
> df
  D  Y        pi Y0 Y1
1 1 15 0.2857143 15 15
2 0 15 0.2857143 15 15
3 0 20 0.2857143 20 20
4 0 20 0.2857143 20 20
5 0 10 0.2857143 10 10
6 0 15 0.2857143 15 15
7 1 30 0.2857143 30 30
```

Consider all the ways treatment could be assigned: (this is what `ri` is doing for us)

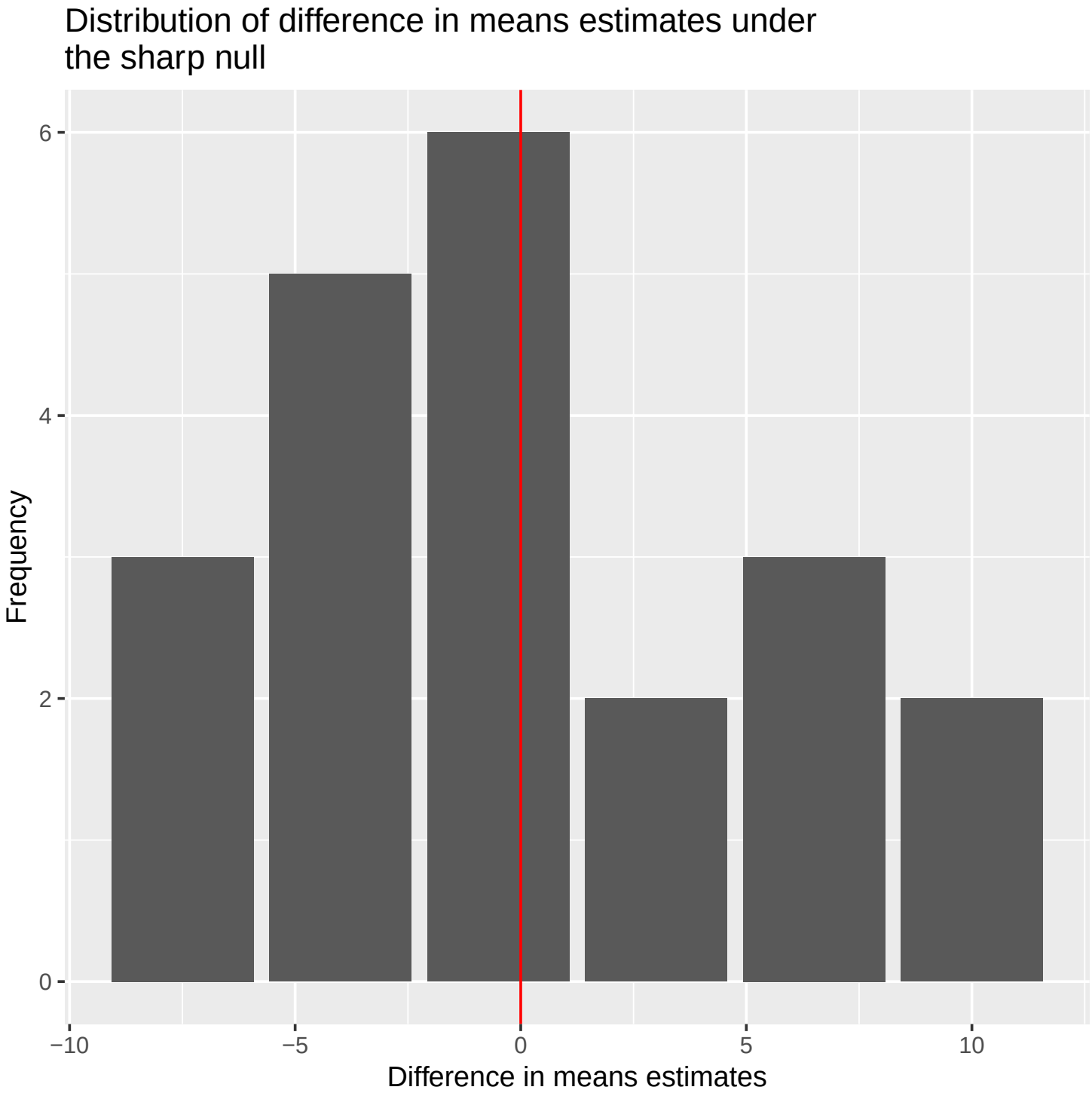
```
> (perms <- genperms(df$D))
  [,1] [,2] [,3] [,4] [,5] [,6] [,7] [,8] [,9] [,10] [,11] [,12] [,13] [,14]
1    1    1    1    1    1    1    0    0    0     0     0     0     0     0
2    1    0    0    0    0    0    1    1    1     1     1     0     0     0
3    0    1    0    0    0    0    1    0    0     0     0     1     1     1
4    0    0    1    0    0    0    0    1    0     0     0     1     0     0
5    0    0    0    1    0    0    0    0    1     0     0     0     1     0
6    0    0    0    0    1    0    0    0    0     1     0     0     0     1
7    0    0    0    0    0    1    0    0    0     0     1     0     0     0
  [,15] [,16] [,17] [,18] [,19] [,20] [,21]
1     0     0     0     0     0     0     0
2     0     0     0     0     0     0     0
3     1     0     0     0     0     0     0
4     0     1     1     1     0     0     0
5     0     1     0     0     1     1     0
6     0     0     1     0     1     0     1
7     1     0     0     1     0     1     1
```

Then generate the sampling distribution of the ATE estimate under the sharp null of no effect.

```
> Ys_null <- list(
+         Y0 = df$Y0,
+         Y1 = df$Y1
+ )
> dm <- gendist(Ys_null,
+               perms,
+               prob=df$pi)
> dm
 [1] -4.0 -0.5 -0.5 -7.5 -4.0  6.5 -0.5 -0.5 -7.5 -4.0  6.5  3.0 -4.0 -0.5 10.0
[16] -4.0 -0.5 10.0 -7.5  3.0  6.5
```

The mean under our null distribution is *exactly* zero. Why?

```
> mean(dm)
[1] 0
```


Distribution of difference in means estimates under the sharp null
Frequency
6
4
2
0
−10
−5
0
5
10
Difference in means estimates

```
> prop.table(table(dm))
dm
     -7.5        -4      -0.5         3       6.5        10
0.1428571 0.2380952 0.2857143 0.0952381 0.1428571 0.0952381
> (pval <- mean(abs(dm) >= abs(dm_hat)))
[1] 0.3809524
```

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- Under the null distribution, if we were to re-randomize the experiment many times, we would see a value *at least as extreme as our estimate* 38.1% of the time.

p-value

- Under the null distribution, if we were to re-randomize the experiment many times, we would see a value *at least as extreme as our estimate* 38.1% of the time.
- That doesn't seem **that** unlikely–more than one in three times, we would see an estimate as large as we got, just by chance.

`ri` produces the same exact p-value.

```
> dispdist(distout = dm,
+          ate = dm_hat,
+          display.plot = FALSE)$two.tailed.p.value.abs
[1] 0.3809524
```

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```

That's because it is doing the exact same thing under the hood.

- Why is this test called an “exact” test?

- Why is this test called an “exact” test?
- Because we know the *exact* distribution of our estimate under the specified null. We do not have to approximate the distribution.
- This will not be true for all of our hypothesis testing…

Hypotheses

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- In our case, we did not find strong evidence to reject the null hypothesis, i.e., our data is consistent with what we would see if the null hypothesis were true.
- Note that we do NOT say that we reject or accept the alternative hypothesis.
- We can only say that our results were not consistent with or were consistent with what we would have seen under the null–i.e., we have evidence to reject or fail to reject the null.

Distributions of estimators

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- Just like other random variables have distributions to describe them, estimators also have distributions.
- We don't know the *true* distribution of the difference in means estimator, for the same reason that we don't know individual treatment effects. (FPoCI!)
- But we DO know what the distribution would be under the null.

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- a defined population
- a defined treatment
- a defined outcome
- a known treatment assignment mechanism
- a given estimating procedure

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The null

- How do we determine what the null is?
- We formalize our hypotheses in terms of the effect we are trying to find in the data. Is there a treatment effect? Is there a difference between these two groups?
- The null is (often, but not always) the case when there is no effect, or no difference.
- We can imagine other kinds of hypotheses, for example that effects are bounded away from zero and positive, or exactly .2. And we can characterize the distribution of our test statistic under the null.

The null

- Is the sharp null of no individual level effects plausible in this setting?

The null

- Is the sharp null of no individual level effects plausible in this setting?
- Does it matter?

Worked example: Fisher's "Lady tasting tea"

Fisher (1935), Chapter 2
Randomization inference by counting assignments

Setup

- 8 cups of tea

Setup

- 8 cups of tea
- 4 cups prepared **milk-first** and 4 prepared **tea-first**

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- The experimenter **randomizes** which cups are which
- The lady claims she can identify which cups were prepared milk-first
- She must label **exactly 4** cups as milk-first (so guessing has a known baseline)

Null + test statistic

- Null (no skill): her answers are unrelated to the randomized assignment

Null + test statistic

- Null (no skill): her answers are unrelated to the randomized assignment
- Encode the randomization as $\boldsymbol{Z} \in \{0, 1\}^8$, where $Z_i = 1$ if cup i is milk-first and $\sum_i Z_i = 4$

Null + test statistic

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- Encode the randomization as $\boldsymbol{Z} \in \{0,1\}^8$, where $Z_i = 1$ if cup i is milk-first and $\sum_i Z_i = 4$
- Encode her guess as $\boldsymbol{G} \in \{0,1\}^8$, where $G_i = 1$ if she says cup i is milk-first and $\sum_i G_i = 4$

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- Encode her guess as $\boldsymbol{G} \in \{0,1\}^8$, where $G_i = 1$ if she says cup i is milk-first and $\sum_i G_i = 4$
- Test statistic (a dot product / contrast):

$$T = \boldsymbol{Z}^\top \boldsymbol{G} = \sum_{i=1}^{8} Z_i G_i$$

which equals the number of milk-first cups she correctly identifies.

Step 1: Observed data (one realization)

```
> tea <- data.frame(
+   cup = 1:8,
+   # randomized assignment: which cups are milk-first?
+   Z = c(1, 1, 0, 0, 1, 1, 0, 0),
+   # her guess: which cups does she label milk-first?
+   G = c(1, 1, 1, 0, 1, 0, 0, 0)
+ )
> tea$correct <- tea$Z * tea$G
> tea
  cup Z G correct
1   1 1 1       1
2   2 1 1       1
3   3 0 1       0
4   4 0 0       0
5   5 1 1       1
6   6 1 0       0
7   7 0 0       0
8   8 0 0       0
> (T_obs <- sum(tea$correct))
[1] 3
```

Step 2: How many assignments are possible?

Under the design, we choose which 4 of the 8 cups are milk-first:

$$\binom{8}{4} = 70$$

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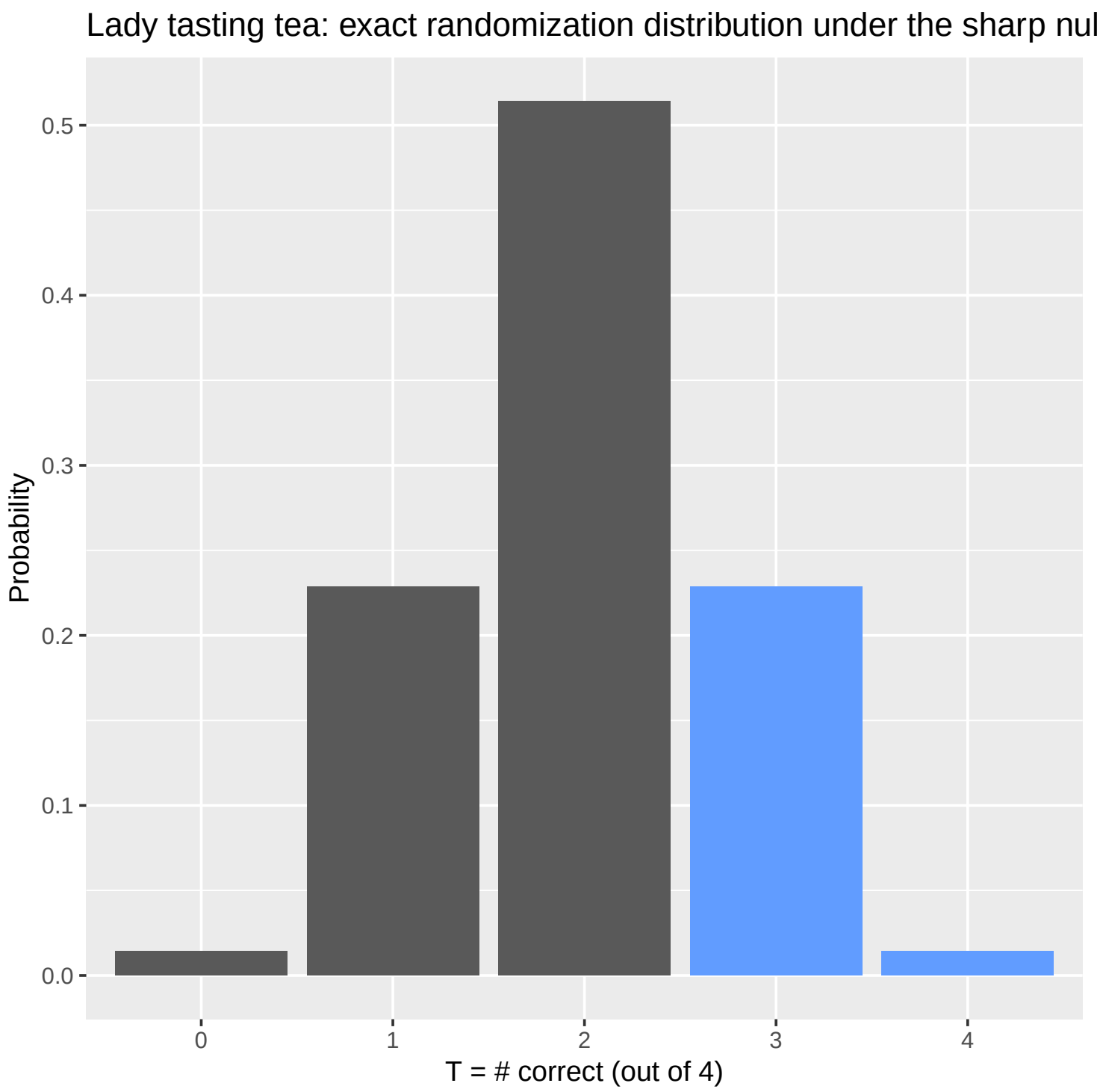
Each assignment is equally likely under the randomization.

Or generate permutations

```
> (perms <- genperms(tea$Z))
  [,1] [,2] [,3] [,4] [,5] [,6] [,7] [,8] [,9] [,10] [,11] [,12] [,13] [,14]
1    1    1    1    1    1    1    1    1    1     1     1     1     1     1
2    1    1    1    1    1    1    1    1    1     1     1     1     1     1
3    1    1    1    1    1    0    0    0    0     0     0     0     0     0
4    1    0    0    0    0    1    1    1    1     0     0     0     0     0
5    0    1    0    0    0    1    0    0    0     1     1     1     0     0
6    0    0    1    0    0    0    1    0    0     1     0     0     1     1
7    0    0    0    1    0    0    0    1    0     0     1     0     1     0
8    0    0    0    0    1    0    0    0    1     0     0     1     0     1
  [,15] [,16] [,17] [,18] [,19] [,20] [,21] [,22] [,23] [,24] [,25] [,26] [,27]
1     1     1     1     1     1     1     1     1     1     1     1     1     1
2     1     0     0     0     0     0     0     0     0     0     0     0     0
3     0     1     1     1     1     1     1     1     1     1     1     0     0
4     0     1     1     1     1     0     0     0     0     0     0     1     1
5     0     1     0     0     0     1     1     1     0     0     0     1     1
6     0     0     1     0     0     1     0     0     1     1     0     1     0
7     1     0     0     1     0     0     1     0     1     0     1     0     1
8     1     0     0     0     1     0     0     1     0     1     1     0     0
  [,28] [,29] [,30] [,31] [,32] [,33] [,34] [,35] [,36] [,37] [,38] [,39] [,40]
1     1     1     1     1     1     1     1     1     0     0     0     0     0
2     0     0     0     0     0     0     0     0     1     1     1     1     1
3     0     0     0     0     0     0     0     0     1     1     1     1     1
4     1     1     1     1     0     0     0     0     1     1     1     1     0
5     1     0     0     0     1     1     1     0     1     0     0     0     1
6     0     1     1     0     1     1     0     1     0     1     0     0     1
7     0     1     0     1     1     0     1     1     0     0     1     0     0
8     1     0     1     1     0     1     1     1     0     0     0     1     0
  [,41] [,42] [,43] [,44] [,45] [,46] [,47] [,48] [,49] [,50] [,51] [,52] [,53]
1     0     0     0     0     0     0     0     0     0     0     0     0     0
2     1     1     1     1     1     1     1     1     1     1     1     1     1
3     1     1     1     1     1     0     0     0     0     0     0     0     0
4     0     0     0     0     0     1     1     1     1     1     1     0     0
5     1     1     0     0     0     1     1     1     0     0     0     1     1
6     0     0     1     1     0     1     0     0     1     1     0     1     1
7     1     0     1     0     1     0     1     0     1     0     1     1     0
8     0     1     0     1     1     0     0     1     0     1     1     0     1
```

Step 3: Exact randomization distribution under the null

```
> dist_df <- data.frame(
+   T = 0:4,
+   prob = dhyper(0:4, m = 4, n = 4, k = 4)
+ )
> dist_df
  T       prob
1 0 0.01428571
2 1 0.22857143
3 2 0.51428571
4 3 0.22857143
5 4 0.01428571
> (p_ge_Tobs <- sum(dist_df$prob[dist_df$T >= T_obs]))
[1] 0.2428571
```


Lady tasting tea: exact randomization distribution under the sharp nul
Probability
0.0
0.1
0.2
0.3
0.4
0.5
0
1
2
3
4
T = # correct (out of 4)

Step 5: Compute and interpret the p-value

In this example, $T_{\text{obs}} = 3$. The one-sided p-value is:

$$p = \Pr[T \geq T_{\text{obs}}] = 0.2429.$$

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- This p-value is computed by **counting assignments** consistent with results at least as good as what we saw.

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- This p-value is computed by **counting assignments** consistent with results at least as good as what we saw.
- It is **not** $P(H_0 \mid \text{data})$ (Fisher rejects "inverse probability" as the general justification for scientific conclusions).

References I

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